Currency Strength: Definition and Basic Properties

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Abstract

In the foreign exchange market, the value of a currency is determined relatively in each exchange market with another currency, and all currencies are interacting strongly, so that it is generally difficult to estimate its solo value. In this study, we introduce a definition of currency strength that quantifies the solo value of the currency. We observe the change of currency strength in big events such as a governmental invention.

Keyword: Foreign Exchange Market, Currency Intervention, Geometric Mean

1. Introduction

In the foreign exchange market, it is observed that there is a triangle arbitrage that generates profit with no risk by exchanging three currencies simultaneously [1]. Such deals cause strong interaction among currency exchange rates.

In this study, using exchange rates and its deal volumes, we define a currency strength that quantifies a solo value of each.

2. Methods & Data

There are many approaches to quantify the value of currency, for example, the effective exchange rate was suggested by combination of exchange rates with other currencies [2]. In this study, we define currency strength by using geometric mean of exchange rates by weights of deal volumes and taking their logarithm in order to take into account the intuition that a stronger currency may be more frequently exchanged. We observe the currency strengths of major currencies, especially focusing on the special day, 31th October 2010 when the Bank of Japan managed the largest currency intervention.

3. Results

We show the time series of currency strength on 31th October 2010 in Figure 1. We find how intervention worked between Yen and USD and also we can find side-effect to other currencies.

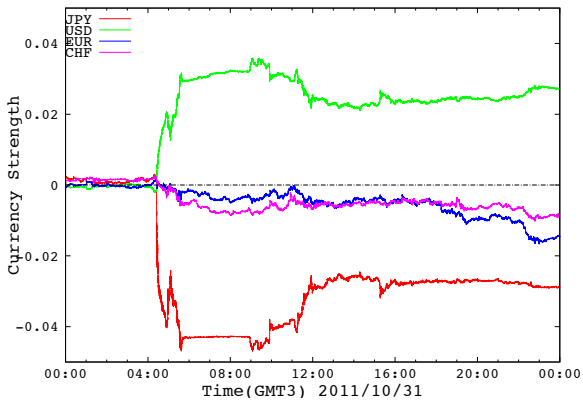


Figure 1. Currency Strength on 31th Octover 2011

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